

**1993
International Amiga
Developers
Conference Notes**

 **Commodore**

Table of Contents

1993 International Amiga Developers Conference Notes

System Software

V39 and AA Compatibility	1
ARexx	27
Overview of V39 Graphics	43
Intuition 3.0	55
Datatypes	77
Changes in the V39 OS	105
IFF	143
3.0 AmigaGuide	159

Hardware

68060: The Fourth Generation in a Heritage	181
Advanced Exec and CPU Issues	185
An Overview of the Advanced Amiga Architecture and Other Future Directions	195
A1200 Hardware Developer Notes & A600/A1200 PCMCIA Slot	219
A4000/A3000 Hardware Developer Notes	257

Japan

Localizing Software for Japan	367
Amiga Application Distribution in Japan	383

Future Directions

Amiga User Interface Directions	389
Retargetable Graphics Specification	419

Multimedia

Introduction to CAMD	439
DSP on the Amiga	489
MediaDevices	515
MPEG and the Amiga/CDTV	535
Amiga Vision Professional	555

Networks

AS225 and SANA II	565
Envoy	595
An Introduction to Fuzzy Logic and Neural Networks	635

Product Development

Object Oriented Programming	647
Product Testing	665
Using Amiga Debugging Tools	669
Writing OS-Friendly Games	691
CDTV Programming and Spooly Device	699
ISO-CD and OPTCD & SimCD	713

ISOCD and OptCD & SimCD

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ISOCd and OptCD

Table of Contents

Overview and Usage philosophy	4
CDTV.TM	4
Memory/resource usage	5
Examples of use	5
General procedures	5
Philosophy	6
System Setup	6
Options	6
ISO	6
Dir Order/Rev	6
Dir Group/Rev	7
Volume ID, Set, Pub, Prep, App	7
PVD	7
Base Sector	7
Split File	7
CDFS	8
Data and Dir Caches	8
File Lock, File Handle	8
Retries	8
Direct Read	8
Trademark	9
Fast Search	9
Speed Independent	9
CDFS Options	10
Restore Defaults	10
Load/Save Layout	12
Examine	12
Optimize	12
Overview	12
Statistics	12
Directory No Cache, Directory Cache	13
Block No Cache, Block Cache	13
File	13
Low Level Read	13
Method	13
Mark What Was Changed	13
Optimize	13
Build	14
Files	14
Source	14
Image	15
Layout	15
Path/File	15
Status	16
Menus	17
Batch	18
Status & General Errors	20
Glossary	21
OptCD	22
Overview	22
CDFS	22
Statistics File Format	23
Batch	24 - 25
General Errors	25
Credits	26

SimCD

Table of Contents

Overview	27
Installation	27
Support for Simulator	28
FakIR	28
Batch Usage	29
General Errors	29
BookFile	29
Batch Usage	29
Examples	30
General Errors	30
DEVS:cdtv.device	32
Commands	33
DEVS:bookmark.device	34
L:CDFS	34
LIBS:playerprefs.library	35
LIBS:debox.library	35
Memory/Resource Usage	36
Once Simulator is Running	36
Files	36
Settings	36
Sim Device	36
Book Mark	36
Card Mark	36
TOC	37
Settings	37
CDFS	37
cdtv.device	37
bookmark.device	38
cardmark.device	38
playerprefs.library	38
debox.library	38
IR	38
No Fast Memory	38
Speed	38
Avail	38
Simulate	39
Load/Save Settings	39
Batch Usage	39
General Errors	39
Glossary	41
Credits	42

ISOCD Overview

ISOCD will examine a directory or partition and create an ISO-9660 image, ready to be mastered into a CDTV disc. The ISO-9660 image will have the necessary data to be used on ISO-9660 and the disc produced can be used on other systems, aside from the Amiga, such as Macintosh, Unix, or IBM compatible computers. Besides CDTV, the disc is usable by the existing CD-ROM file systems for the Amiga. Most importantly, the SimCD simulator can treat a partition created with ISOCD as if it were the CD0: drive of a CDTV.

ISOCD has several important features:

- You can graphically re-arrange the actual order of files, changing how they will be accessed. This will greatly improve the access times on CD-ROM drives.
- These layout files can be re-used.
- It uses a minimal amount of memory for the image creation.
- It allows the setup of custom CDTV CDFS options.
- Files can be grouped and ordered automatically.
- ISOCD can optimize the access order of your files and directories based on information gathered by OptCD. OptCD monitors CD0: access while you are walking your application through its paces.
- The ISO-9660 headers can be produced separately to allow a small initial data track for CD-ROMs that are audio compatible.
- The layout file is text based so it can be manipulated by other tools of your own.

Usage philosophy

You can use ISOCD to make ISO-9660 image files to be sent off and made into CDTV CD-ROMs. With SimCD and a large enough partition on your hard drive, you can use ISOCD to create simulated discs for testing your application without actually pressing discs! Of course, final testing should be done with actual CD-ROMs on a CDTV, but most of the development cycle can occur without the hassle and expense of outside disc pressing. Actual placement of the files and directories on a CD-ROM is very important for maximum performance of your application on the SLOW seeking CD-ROM drives. Because of this, ISOCD allows you to rearrange the order of files and the like. These ordered listings can be saved and loaded as layout files.

CDTV.TM

CDTV.TM is provided with ISOCD as the trademark file for CDTV discs. This file allows the application to run on a CDTV. ISOCD will place this file close to the beginning of the

CD-ROM since it is one of the first files read at boot time. If this file is present in the current directory, it does not need to be present in the source directories. If properly found, it will show up in the examined list as <Trademark>.

Memory/resource usage

Run time memory usage of ISOCD is strictly dependent on the number of files and directories in your application. To figure rough memory usage, take the size of internal buffer, add 140K and an average of 50 bytes per file or directory entry. The internal buffer is normally 256K in size, refer to -f under Batch for changing this size. So an application with 5,000 files and directories and a default buffer would take approximately 650K of memory to build.

Examples of use

To help you get started quickly, let's do a very simple ISO build right now.

Follow these steps:

- Start up ISOCD.
- In the Files box, click on [Source].
- Select the "S:" directory, click [OK].
- In the Files box, click on [Image].
- Enter "RAM:Test.iso," click [OK].
- In the Actions box, click on [Examine].
- In the Actions box, click on [Build].
- That's it, "RAM:Test.iso" is an ISO-9660 image ready to be made into a disc.

General procedures

Usage of ISOCD is rather simple. The following steps represent a simple build:

- Check that the options are set correctly.
- Set the Source directory.
- Set the Image to a file or partition.
- Examine.
- Rearrange files as desired.
- Save the layout file if desired.
- Build.

After an examine, you can save the list of files out as a layout file. This file can be externally edited as to the order of the files and reloaded or you can use the editor within ISOCD. Later usage on the same project can use the layout file so that you do not have to re-order anything but the new files.

Philosophy

The ability to re-arrange the layout of files in the image is important to optimizing the execution timing of an application. The directory structure of the source material is always preserved, but the physical layout of files is completely independent. This means that the files used first on booting should probably be at the head of the list, closer to the center of the CD-ROM. This minimizes the deadly seek time inherent to CD-ROM technology. If the files are arranged properly, a great number of seconds can be shaved from the load time of your title. A little thought on the physical layout of files can actually create a responsive title - one that actually fits the needs of the consumer.

System Setup

ISO-9660 images or partitions can be very large. It is wise to have more than twice as much hard drive space as your application actually uses. It is even more wise, in fact almost Guru thinking, to use a separate drive for the image. This reduces the risk involved with such large projects. ISOCD is very meager with its memory usage, but if there are a tremendous number of source files, you should have at least 2 megabytes of memory.

In case you feel overloaded with all of the files needed for a multimedia application, consider Hypermedia's Fred Fish CD-ROM. The version 1.5 disc created had almost 600 megabytes and 67,000+ files! ISOCD built the image in just a little over two hours. The Fred Fish Online CD-ROM was 415 megabytes and 4,000 files. It took ISOCD 16 minutes. Of course, your performance will vary according to disk speeds, DOS fragmentation, etc.

Options

Options are saved with the layout file. They control the examine process, the header of the ISO-9660 image, and the interface to the CDFS in CDTV during the boot process. Directory ordering and grouping must be set before an examine, and the other options must be set before a build.

ISO

ISO-9660 provides an area in the header of an image called the Primary Volume Descriptor, or PVD. Here you specify the title, publisher, etc.

Dir Order/Rev

The directory order present on the original source is always preserved. ISO-9660 specifies the ordering of files and directories to be alphabetical. But we are free to place the physical locations of the files and directory entries. When examining and building the ordered list of files, ISOCD can perform automatic arrangement of files. Under Dir Order, you can specify

file arrangement within a directory alphabetically, by size, date, or how they are originally ordered. Order is from A-Z, least to largest, or oldest to most recent. This can be reversed. Of course, you are free to arrange the files after an examine.

Remember that this is the physical position on the CD-ROM, not the order "LIST #?" would produce, since LIST uses the ISO-9660 directory structure to report files. Also note that these orderings are per directory, not over the entire CD-ROM.

Dir Group/Rev

In conjunction with ordering, you can group files together within a directory. You can group ".Info" files, directories, common file extensions, or not at all. Grouping directories is the most useful since the CDFS uses a directory cache system that may be able to load all of the current directories in one read if they are grouped together. This is obviously application dependent.

Volume ID

Volume ID is the CD-ROM title, 32 characters maximum. The remaining descriptors are limited to 128 characters.

Volume Set

Place the volume set name here.

Volume Pub

The publisher is specified here.

Volume Prep

Indicates the actual preparer of the data.

Volume App

Indicates the application name.

PVD

Though only one PVD is needed for a CD-ROM, you can specify more than one for data redundancy to improve reliability. CDTV will try another PVD if there is an error reading one.

Base Sector

Split File

Base Sector and Split File provide the ability to place the bulk of the application in a different track than the first. You specify a position for the data track in the Base Sector and a file for the header. This gives you two data tracks, one which is the header and must be placed at the beginning of the CD-ROM, and another that can be placed after audio tracks. This renders a CD-ROM that the consumer can use as an audio CD and it really is a CDTV disc as well!

CDFS

The CDFS in CDTV allows custom configuration of many variables through a data area in the PVD. You can set them here to tailor CDTV to your application.

Data Cache

CDFS has a cache for reads, specified in number of sectors. Keep in mind that each sector cached represents 2048 bytes of memory. There can be from 1 to 127 sectors cached. The default value is 8 sectors.

Dir Cache

More important, CDFS caches directories. You can specify from 2 to 127 sectors to be cached, each using 2048 bytes. On a CD-ROM, there is always seek timing to be considered. If all directory entries for an application can be read into the cache in one read, the savings in access time is tremendous. Properly used, the CD-ROM can be almost as responsive as a hard disk. Again, this uses memory that cannot be recovered by the application. The default value is 16 sectors.

File Lock

File locks are managed in pools, from 1 to 9999 locks, each using 32 bytes. The default value is 40 locks.

File Handle

File handles are also managed in pools, from 1 to 9999 handles, each using 24 bytes. The default value is 16 handles.

Retries

You can specify from 0 to 9999 retry attempts on read failures. The default value is 32.

Direct Read

This option allows an application to wring the ultimate in performance out of the original CDTV. CDTV has a bug in the CD-ROM drive hardware that occasionally will not transfer the last few bytes. The system handles this for all applications, but at a performance penalty. There are three ways to read data at the optimal, unprotected, rate:

- Use the `cdtv.device` directly with the `CDTV_READ` or `CDTV_READXL` commands.
- Turn on Direct Read for an opened file by sending a `ACTION_DIRECT_READ` packet for the file.
- Turn on Direct Read for the entire CD-ROM, with this option.

When reading directly, you must pad your buffers and your disc data by the value `READ_PAD_BYTES`, and request `READ_PAD_BYTES` more bytes than you would otherwise. Thus, any missing bytes are part of the padding, and ignored. `ACTION_DIRECT_READ` is defined in the CDTV manual. `READ_PAD_BYTES` is defined in `<devices/cdtv.h>`.

You must understand that normal workbench commands, `LoadSeg`, and other programs *do not* know to handle their reads differently with Direct Read on for the entire CD-ROM. This means writing your own custom loader and executing only your programs on a CD-ROM with Direct Read set. This also means, for most practical purposes, that this option is not really usable. We still *highly* recommend using `CDXL` (`CDTV_READXL`) and direct `CDTV_READs` within your application in any case. It is easier and a lot faster!

Any new CDTV machines or CD-ROM drives will not suffer under this curse.

Trademark

The Trademark file is necessary for the application CD-ROM to boot under CDTV. It is removed from the screen with `RMTM` on booting, please refer to the CDTV documentation. `ISOCD` places this file outside of the directory at the very inside of the CD-ROM for faster booting. The file `CDTV.TM` is the Trademark file. If you place it in the current directory when you run `ISOCD`, it does not need to be in the source directories. This option defaults to on.

Fast Search

Fast Search is an optimization available if the volume is prepared with directories that are sorted case insensitive. `ISOCD` works this way so it is normally on. This optimization enables `CDFS` to stop its search for a file in a directory if it passes the position alphabetically.

Speed Independent

CDTV uses a CD-ROM drive that reads in data at 75 sectors a second, or 150K/second. As we all know, faster drives are being produced. This option lets any future machine know that the application would be more than happy with faster reads. This is up to you, we will not twist your arm.

Of course, all of the caveats of good programming apply if you want to be speed independent. The 75hz timer within CDTV will always be available for easy timing control, independent of NTSC or PAL. We recommend you use this, it is `CDTV_FRAMECALL` with `cdtv.device`.

This option defaults to off.

CDFS Options

Option	Default	Minimum	Maximum
Data Cache	8	1	127
Dir Cache	16	2	127
File Lock	40	1	9999
File Handle	16	1	9999
Retries	32	0	9999
Direct Read	OFF		
Trademark	ON		
Fast Search	ON		
Speed Ind.	ON		

Restore Defaults

This resets all of the CDFS options to their defaults. It is available for those times when real confusion sets in.

Load/Save Layout

Layout files contain all of the information necessary to build an ISO-9660 mage. Even though ISOCD lets you edit this list, it can also be edited on the outside by any TEXT editor, not a word processor, unless it can save as a text file. ISOCD will not understand any particular word processor format or special codes.

Here is an example layout file. It's a 1.3 devs directory examined using alpha order and dir grouping.

0 0 1 0	F0003 gb
0 0 1 0	F0003 i
2 1024	F0003 is
8 16 40 16 32	F0003 n
0 1 1 0	F0003 s
CDTV_TEST	F0003 usa0
Test_Set 1 of 4	F0003 usa1
Pantaray, Inc.	F0003 usa2
Kenneth Yeast	F0004 Alphacom_Alphapro_101
Application Name	F0004 Brother_HR-15XL
TestHeader.iso	F0004 CalComp_ColorMaster
Test.iso	F0004 CalComp_ColorMaster2
0004 System1.3:devs	F0004 Canon_PJ-1080A
0001 <Root Dir>	F0004 CBM_MPS1000
0001 clipboards	F0004 Diablo_630
0001 keymaps	F0004 Diablo_Advantage_D25
0001 printers	F0004 Diablo_C-150
	F0004 EpsonQ
H0000 <ISO Header>	F0004 EpsonXOld
P0000 <Path Table>	F0004 EpsonX[CBM_MPS-1250
P0000 <Path Table>	F0004 generic
C0000 <Trademark>	F0004 Howtek_Pixelmaster
D0001 <Root Dir>	F0004 HP_DeskJet
D0002 clipboards	F0004 HP_LaserJet
D0003 keymaps	F0004 HP_PaintJet
D0004 printers	F0004 HP_ThinkJet
F0001 clipboard.device	F0004 ImagewriterII
F0001 kickstart	F0004 Nec_Pinwriter
F0001 MountList	F0004 Okidata_293I
F0001 narrator.device	F0004 Okidata_92
F0001 parallel.device	F0004 Okimate_20
F0001 printer.device	F0004 Quadram_QuadJet
F0001 ramdrive.device	F0004 Qume_LetterPro_20
F0001 serial.device	F0004 Toshiba_P351C
F0001 system-configuration	F0004 Toshiba_P351SX
F0003 cdn	F0004 Xerox_4020
F0003 chl	E0000 65536
F0003 ch2	
F0003 d	
F0003 dk	
F0003 e	
F0003 f	

The layout file is a text file that contains the options, directory structure, and editable order list. The first two parts of the file are *not* to be changed, this would produce unpredictable results. The first part of the file is 11 lines of option data. The next section is a listing of the directory structure, terminated with a blank line. The third section is yours to rearrange, it is the order of files/dirs/etc., on the CD-ROM.

The actual text in each of the lines should *not* be changed, only the order of the text lines is changeable. Also, no line should be deleted or added. This should only occur within ISOCD.

Once you have your order figured out and saved in a layout file, you can automate the build process during development. Remember to specify the image file or partition before saving the layout file. Assuming your layout file is called Test.lay, simply call ISOCD as follows:

```
ISOCD -lTest.lay -b
```

This would build an image without any interaction, allowing its usage in an arexx script or batch file.

Examine

This will examine the source directory and build the list necessary to create an ISO-9660 image. You can alternately load a layout file from a previous examine. The examine will take into account the directory order and grouping specified in the options. The examine will display each directory that it is loading and you can abort at any time. An examine will clear any previous list in memory.

After an examine, Status will report the number of bytes in the image, the number of directories, and the number of files. This number of bytes is usable in creating a TOC file, assuming you have audio tracks as well.

The ISO-9660 standard specifies that directories cannot be greater than eight deep. ISOCD will allow any number deep, after all - this is the Amiga - but will report if you have violated the standard.

Optimize

Optimize will take data accumulated in a statistics file by OptCD and use the totals to change the order of files and directories in your layout. Despite the type of data collected, you have the choice of what will be used in optimizing the list. If you wish to re-arrange only directories, turn on the Directory No Cache and Cache only. If you wish to optimize files only, use File only. Normally you will have no need for Block or Low Level Read, but it is here for completeness.

You can only optimize a layout once, since the statistics file is based on the existing CD layout only. If you wish to try again, simply load the layout file again.

Note that all reads that are used are given equal priority. You may wish to optimize directories only at first, then optimize files after a rebuild and another OptCD monitoring.

Statistics

This is the file produced by OptCD when monitoring your application. See OptCD. It provides the access data and is only valid when paired with the layout that exactly matches the CD0: image.

Directory No Cache

Directory Cache

Note directory reads that were or were not in the directory cache.

Block No Cache

Block Cache

File reads and directory reads that are not in the directory cache, use the block reads. This notes those block reads in or out of the cache.

File

All file reads will be noted.

Low Level Read

Note any read of the CD.

Method

Float Most Read to Top

This will move the most read to the top after <Root Dir>.

Group Entries by Access Order

This method is more complicated. It notes when one file or directory is read after another. Multiple occurrences of a pairing give it a higher priority. The highest pairings are done first, then the next, and so on. If an entry is to be moved after another and it is already paired, it will not be moved. However, if it has one paired with it, they both will be moved, etc. It is important that you walk through your application with the order in mind when you plan to use this method.

This method works best when used for files only.

Float Dirs to Top - No File

This is provided if you wish to simply move all directories after <Root Dir> at the top. If you have just a few directories, you may wish to fit all of them into the directory cache and never require a CD read for directories ever again, see CDFS under Options.

Mark What Was Changed

This will highlight the files or directories moved during optimize. In this way you can see what was changed.

Optimize

Perform the optimization. Remember that you can try another method by loading the layout file again.

Build

Build actually creates the ISO-9660 image file or partition. Remember that an image file can be very large. If an image is written to a partition, then that partition can then be simulated as CD0: or as just an ISO-9660 volume with SimCD. Refer to the SimCD documentation on the usage of CDFS and the simulator. While building, ISOCDB will display a progress bar and can be aborted at any time.

The progress bar does not show writing of the directory blocks, so if you have all of your directories at the beginning of a list, there will be a small delay before the progress bar registers.

Warning - Warning - Warning

Please exercise unbelievable caution when using build on a partition. Once started, it will obliterate anything that was previously on that partition. Use it as you would use the DOS format command.

When you specify an image, you can enter a file name or a partition, such as CDN:. Because the file requester is dual purpose, entering a partition that is not an AmigaDos partition in the Drawer text will cause the requester to attempt reading of the partition. Of course, since it is not able to read the files in that partition, the requester will report an error. Simply choose Cancel, then OK in the requestor. Or, as we have found, enter the partition name, such as CDN:, in the File text.

If you are using the simulator, remember to write to the partition that is being simulated, such as CDN:. CD0: is a read only partition based on the ISO-9660 partition and cannot be written over. ISOCDB will diskchange the partition being built and CD0: if it exists, but ISOCDB does not know about any other devices that might be simulated as well from the image partition. You must remember to diskchange these yourself.

Files

Source

This sets the source directory to be examined. If you have previously examined, changing the source will clear the list. All files and directories in the source will be examined.

If you do not specify a source directory on the command line, it will default to the current directory. Remember that the layout file stores all of the information, so loading a layout file will set the source directory to the one for that list.

Image

When you specify an image, you can enter a file name or a partition, such as CDN:. Because the file requester is dual purpose, entering a partition that is not an AmigaDos partition in the Drawer text will cause the requester to attempt reading of the partition. Of course, since it is not able to read the files in that partition, the requester will report an error. Simply choose Cancel, then OK in the requestor. Or, as we have found, enter the partition name, such as CDN:, in the File text.

Layout

This sets the layout file name, refer to Load/Save Layout for more information.

Path/File

This gadget shows you the list of all entries to be written to the ISO-9660 image. It also allows you to re-arrange the order of everything, except for the ISO-9660 headers, always at the beginning, and empty space at the end of the image.

To move a file or set of files: first click on all of those to be moved, then, while holding down the Ctrl key, click where you want them to go. ISOCD will move the marked entries after the entry clicked on with Ctrl. If you wish to use Shift or Alt, instead of Ctrl, specify -q1 for Shift, or -q2 for Alt on the command line. You can also select an entire range by simply holding down the mouse and moving over the entries. The list will scroll for you, accelerating if the mouse continues moving outside the gadget. To deselect an entry, click on it again; extended select works in reverse as well.

ISO-9660 deals with everything in sectors of 2048 bytes. All files consume an even number of sectors; this means a file of 920 bytes uses 2048 bytes on the CD-ROM. This applies to all items on a CD-ROM, such as Path tables or directories. ISOCD displays the file sizes as they originally are, or there is a menu option that lets you display the sizes rounded to the sectors occupied or even dates instead. Amiga-S sets ISOCD to display actual sizes, Amiga-E sets to rounded sizes, and Amiga-D sets to display dates.

Dates are displayed like the LIST command, except for "today," etc.

The Path number is an ISO-9660 convention. For entries such as Path Table or Empty Space, they have no meaning. But for a directory, it is the directory number and for files, the directory they belong under. This preserves the directory structure since ISOCD internally remembers what directory a directory belongs under. In other words, if a given directory has a path number of 12, any file with a path of 12 is under that directory.

Besides directories and files, there are five other entry types.

- The ISO-9660 Header is shown for completeness, since you cannot move it. It encapsulates the 16 blank sectors, the PVDs, and the terminating volume descriptor.
- The Path Tables provide a short cut for accessing the CD-ROM. A second one is provided for reverse byte systems, such as Intel based computers. This can be removed by -p on the command line. We must caution you that many disc mastering systems require the extra path table in order to press the CD-ROMs.
- The Trademark file is necessary for your application to run on a CDTV. It does not need to be in the list of files in your source directory if it is in the current directory. The Trademark file is called CDTV.TM and is provided with ISOCD.
- The Root Dir is the root directory of the CD-ROM.
- Empty Space is provided at the end of the volume because older CDTV units could experience problems if someone attempted reading past the end of a CD-ROM. Empty Spaces can also be added to other places, refer to Add/Remove Spaces below.

You will notice that the Path Tables, Trademark, and Root Dir are placed at the beginning of the list. This improves boot time and running performance. It is also a wonderful idea to hand place the files used at boot time in order of usage at the beginning of the list, after any directories that need to be cached at the beginning. This takes a certain amount of thinking about how your application starts and might lead you to try some different philosophies. What works on a hard drive, such as echo commands and the like, are ridiculous on a CD-ROM. Consider just one program in your startup-sequence that handles everything possible. Remember that CD-ROMs are not changeable by the user, so a lot of the methods that are necessary for a changing environment such as a hard disk, are not useful on a CD-ROM.

The layout file will contain your arrangement of files so that you do not have to re-arrange entries again. If you are loading an older layout file for a source directory and there are new files or some have been deleted, ISOCD will compensate. New files will be placed at the end, you will need to remember to move them to where you would like. The actual directory structure cannot change, as this would change all of the path numbers, described above. If you have added or removed any directories, you will need to examine again.

The layout file can be externally edited by your favorite text editor if you wish. Refer to Load/Save Layout above for more information.

Status

The first line reports the size of the ISO-9660 image, number of directories, and number of files. The second line reports any usage errors, refer to Status Errors.

Menus

- **Amiga-? "About..."**
About box.
- **Amiga-Q "Quit"**
Quit the program. It is also available on some of the little windows to leave the window.
- **Amiga-S "Display Sizes"**
Display the size of entries, such as file sizes or directory sizes.
- **Amiga-E "Display Sizes Rounded"**
Round the entry sizes to the actual space occupied on the CD-ROM, since the CD-ROM uses sectors of 2048 bytes.
- **Amiga-D "Display Dates"**
Display dates in the LIST format instead of sizes.
- **Amiga-F "Find File..."**
This allows you to jump to a given entry in the list, useful if you have a large set of files. It uses the AmigaDos search pattern, case insensitive, just like dir. Next and Prev is available to continue the search.
- **Menu items available:**
 - Amiga-N "Next"
 - Amiga-P "Prev"
 - Amiga-H "Help"
 - Amiga-Q "Done"
- **Amiga-A "Add Spaces..."**
ISOCD provides a special capability for future use by allowing blank spaces within an ISO-9660 image. Adding spaces after certain files would allow someone to update just that file within an image being used for simulation. In this way, someone could just update the size field in the directory and rewrite a new version of a file instead of going through an entire build. A future utility might just provide this capability.
leave the window.

To add spaces, simply mark those files that need spaces, choose this option, and specify the sizes. If no files were marked, one empty space would be added to the beginning of the list to be moved later. The files marked would be unmarked after an add.
- **Amiga-R "Remove Spaces"**
Likewise, mark those Empty Spaces to be removed, then choose this option. If none is marked, ISOCD will remove the first available, keeping in mind that one is always needed at the end of the CD-ROM.

Batch

USAGE: ISOCD [options] <directory>

- l<file> Layout file
- o<file> ISO-9660 output file or device
- s<file> Split ISO descriptor file
- a<file> Statistics file for Optimize
- b Build image (requires layout file (-l), will exit when done)
NOTE: Use this option with the greatest of care...
- x Verbose debugging information
- f<n> # of sectors in file buffer [128]
- m Use minimum size window
- q<n> Change qualifier for insert in lists [0]
(0=Ctrl, 1=Shift, 2=Alt)
- t Do not use topaz font, use system font instead
- p Do not add Path Table for reverse byte systems
- v Do not append version number ";1" to file names

Options:

<directory>

This is the directory or partition to be examined. If not specified, ISOCD will assume current directory.

-l<file>

Load layout file previously saved by ISOCD. ISOCD will ignore any missing files from the layout file list. Any new files will be appended to the end of the list. The directory structure cannot change however, there cannot be any missing or new directories in the list. This changes the directory numbering scheme of ISO-9660 such that ISOCD would not be able to reconfigure a given list. If directories themselves have changed, re-examine and re-order the list. This option is often used with -b to automatically build after loading the layout list. In this fashion, builds can occur within batch file or arexx scripts, automating program simulations. Layout files specify the build image within the file, so the -o<file> option is ignored.

Example:

ISOCD -lTest.lay -b

-o<file>

This is the output image file or partition to be built. Building to a partition allows you to use SimCD and simulate that partition as CD0: after a build. Remember to diskchange any dependent devices after a build, see Build. Please use great care when writing to a partition. ISOCD will

blindly trust that you are prepared to obliterate that partition with an ISO-9660 image. If the partition has not been used for an ISO-9660 image before, ISOCD will ask before proceeding with the build. But it will ask only once. The build process should be used by professionals, do not try this at home.

-s<file>

ISOCD can create a separate header image file. Please refer to Split File under Options.

-a<file>

Statistics file for Optimize function. This file is created by OptCD while monitoring your application. Refer to Optimize and OptCD.

-b

This will automatically build an image file if you have specified a layout file with -l<file>. This option is commonly used within a batch or arexx script, especially when doing repetitive testing with the simulator. Again, it cannot be stressed enough, please use this with *great care*.

-x

This turns on some primitive and profuse debugging text. If something is not happening correctly, this will help to locate where things go wrong. I assume that you are using enforcer, mungwall, io_torture, etc., for your good health and mine.

-f<n>

ISOCD uses a 256K buffer, 128 - 2K sectors, for its file transfers. Performance can be improved with larger buffers, if desired.

-m

ISOCD asks for a fat window when running so that the file list is as large as possible. -m will use a minimum size window for simpler usage.

-q<n>

When specifying where a group of marked files are to be moved to in the file list, you click where they are to go while holding down the Ctrl key. This allows you to use another modifier key, such as Shift or Alt.

-t

Since the list of files requires a fixed width font, ISOCD uses the Topaz font. If your system font is not too large and is fixed size, specify -t and ISOCD will use it instead.

-p

ISOCD will automatically include a path table entry for the "other" computer systems that use backwards ordering of bytes. This is often necessary for even CDTV specific titles, since many mastering systems are done on these "other" systems and they often need to look at the image. If you really do not want this extra path table, just say no with -p.

-v

In a similar vein, many file systems require the ";1" version strings on all files. CDTV will handle file names with or without these strings, but you can turn them off with -v. Currently there are no provisions for actually specifying other versions instead of ";1".

Status Errors

"Need to Examine first"

A given function can only work if you have done an examine first to provide a list of files/dirs/etc. You cannot build until you have something to build, etc.

"Too many dirs deep for ISO"

This one reports that the source examined happens to be more than eight directories deep. ISO-9660 standard does not accept this, though CDTV and ISOCD will.

"Incompatible dir heirarchy"

When loading a layout file, there cannot be any additional or missing directories. If there are, the load will not complete.

"No more Empty Spaces to remove"

You are attempting to remove an Empty Space when there are none available to remove. There must always be the one at the end of the CD-ROM.

General Errors

"Batch option requires a layout file"

If you specify -b to build automatically, you must also specify -l<file> to load a layout file to be built.

"Not an Amiga system"

Failed to open graphics.library.

"Intuition (V2.0) not available"

ISOCD requires v2.0 of intuition or greater.

“Could not get current directory”

This error should not occur.

“Interface not available”

Probably not enough memory for window.

“Could not create gadgets”

Probably not enough memory for gadgets, or you are using -t and your system font is too large.

“No Memory”

Not enough memory.

Glossary

CD0

The CD-ROM drive on a CDTV or the partition simulated as this drive with SimCD.

CDFS

The CDTV CD File System. This allows you to treat CD0: as just another Amiga drive, even though it is in ISO-9660 format.

cdtv.device

The device that controls the CDTV hardware or the simulation of it with SimCD.

CDTV.TM

The trademark file that allows an application to run on CDTV.

CDXL

The fastest way to transfer data from the CD-ROM drive into the CDTV. CDXL basically starts a read and begins transferring the incoming data into places in memory that you specify. CDXL nodes control the shotgun effect of the data being read in, since the read does *not* pause.

ISO-9660

Refers to the 1988 ISO-9660 standard, specifically ISO 9660: 1988 (E), corrected 1988-09-01.

Layout Files

These text files store the information necessary to build an ISO-9660 image, including the order of files and directories.

PVD

Primary Volume Descriptor. This is the header for the CD-ROM and must always be at the beginning of the CD.

OptCD Overview

OptCD is separate program available to collect information that will allow ISOCD to optimize your disc layout. Every time your application reads from CD0:, be it a directory or a file, OptCD will note the access according to the type and level of the read. This data is used by Optimize in ISOCD to re-arrange the order of the directories and files. You can move the most often accessed files to the beginning of the CD or group together files that are often accessed after each other. The Optimize function of ISOCD takes into account the number of accesses when weighing which file or directory to move, so you should keep this in mind while walking your application through its paces.

A simple walkthrough, assuming you are simulating:

- Run OptCD with a statistics file.
- Walk the application on CD0: through its common usages.
- Turn OptCD OFF.
- Use Optimize in ISOCD to rearrange the entries with the statistics file produced.
- Rebuild image.

You may wish to work through this process several times to get the arrangement you are most happy with. We recommend keeping some of these layout files produced around until you are finished. In any case, you need to keep the final layout file!

You *must* have the layout file that produced the CD0: that you wish to optimize. OptCD records the reads of CD0: as blocks only, so determining which file or directory was read requires an exact knowledge of the source. If you created the image without any rearranging of entries and the source has not been changed, you can get away without a layout file the first time. Of course, once rearranged, only a layout file knows the order.

CDFS

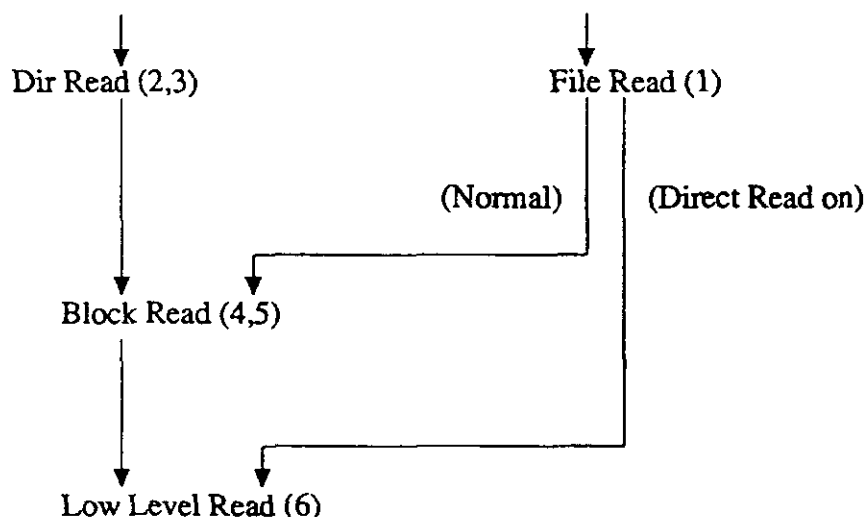
OptCD normally monitors file reads and directory reads, whether they are caught in the cache or not. This gives equal access to all read activity. In this way, multiple iterations of the optimization process should produce similar arrangements for the "most optimal." Cache hits are reads that do not actually read from the drive - but have been previously read into memory. Because part of the monitoring is actually within the file system, CDFS, multiple levels of monitoring were provided. Even though you probably are never going to need to know this information, we'll provide it anyway. You can monitor only actual drive reads, cache reads only, directory reads only, etc. There is even a monitor for all DOS packets if you have anything in particular you wish to parse yourself. With the Optimize function in ISOCD, you select which event type you wish to use, so monitoring anything extra does not hurt here.

CDFS MONITORS

- 0: All DOS Packets
- 1: File Read
- 2: Dir Read (Not Cache)
- 3: Dir Read (Cache)
- 4: Block Read (Not Cache)
- 5: Block Read (Cache)
- 6: Lowest level read

The read monitors (1-6) are layered within CDFS as follows:

CDFS READ STRUCTURE



Briefly, both directory and file reads use the block system, except when Direct Read is on for a file or CD - refer to Direct Read under CDFS. In this case, file reads go directly to the lowest level. Directory reads implement their own caching system and the block level of the reads implement another. Blocks are 2048 byte sectors on the CD0: drive. If you wished to monitor only actual reads to the drive, you could monitor the lowest level only, level 6.

Statistics File Format

The statistics file is a text file with a line for each action. The first number on the line is the code, 0 through 6, for the type of action, e.g. 0=packet, 1=file read, etc. The second number, except for DOS packets, is the block read. DOS packet lines contain the actual three packet

arguments. This is mostly useless information since memory is dynamic, but its provided anyway. Both file read 1 and low level read 6, have a second argument, the length in blocks of the read.

An example listing, monitoring everything:

```
0 8 00000000 07cdc784 ffffffff
0 23 07ca0cac 07cd75f0 00f94dbc
2 32
4 32
6 32 9
0 15 07ca0cac 00000000 00000001
0 8 00000000 07cdc784 ffffffff
0 23 07ca0ccc 07cd75f0 00f94dbc
3 32
```

Batch

USAGE: OptCD [options] <Statistics file> [ON/OFF]

- b<n> buffer size in K (default 8)
- m<#...> events to monitor (default 123 - if not specified)
 - 0: All DOS Packets
 - 1: File Read
 - 2: Dir Read (Not Cache)
 - 3: Dir Read (Cache)
 - 4: Block Read (Not Cache)
 - 5: Block Read (Cache)
 - 6: Lowest level read

[ON/OFF] Force OptCD on or off

Options:

<Statistics file>

The text file of activity from CDFS. If it exists, OptCD will just append more lines. Keep in mind that the size of this is related to the amount of activity, though it is still small.

-b<n>

This specifies the buffer size for messages from CDFS on the activity being monitored. You might want to enlarge this buffer if you have a lot of CDFS activity in a short period of time.

-m<#...>

These are the events that OptCD and CDFS can monitor. If you only wanted to monitor only directory reads, cached or not, you would specify:

OptCD <file> -m23

For an explanation of the events, refer to CDFS above.

[ON/OFF]

The first invocation of OptCD will turn it on and the next will turn it off. ON or OFF allows you to force OptCD on or off, this is useful in arexx or batch files.

General Errors

“Unable to open cdfs.library”

You have not started the simulation of CD0: or you have an older version of the CDFS part of the simulator.

“Cannot install notice”

OptCD could not create a message port named "MP_OptCD.Rendezvous".

“Cannot open Statistics file”

OptCD could not open the file.

“Not enough memory for messages”

There is not enough memory for the buffer. Don't forget to leave enough memory for the application you are going to run!

“Cannot get FSE port”

Is there another CDFS FSE monitor program running besides OptCD?

“Unrecognized message”

Not sure how this message could occur, it indicates a bogus message from CDFS.

“Failed to write statistics file”

Disk may be full.

Credits

Documentation by:

Kenneth Yeast

Software by:

Carl Sassenrath [Alpha version of ISOCD, co-designer and producer]

Kenneth Yeast [Graphical ISOCD, ability to rearrange files, and final version]

Special thanks:

Ben Phister [helped at Commodore to shape ISOCD]

Chris Henry [Hypermedia, provided invaluable beta testing with the Fred Fish
CD-ROMs]

SimCD Overview

CDTV is more than just an Amiga: it is a low cost multimedia player, an Amiga married to a CD-ROM drive, a different beast all around. With CD-ROM as its medium, it brings to the user large amounts of data to make interactivity actually work. We are all just now seeing the real potential of this combination.

Developing for CDTV requires testing with all of this data in its final form, on a CD. This means expensive and slow mastering of test discs. SimCD changes the rules and allows you to create and test on the same development machine, cutting test discs only rarely. SimCD allows you to use a partition on a hard drive as the actual CD for testing. ISOCD is used to prepare this partition as a valid ISO-9660 image.

SimCD also uses the same ROM that is on CDTV, modified for simulation on the Amiga. Read, seek, and CDXL times are simulated, even considering the rotational latency of the CD. The infrared remote is simulated with the keyboard, bookmarks and cardmarks are available, preferences, screen saver, etc. are also simulated.

Because all of this is simulated on your development machine, you can also use enforcer, mungwall, io_torture, etc. This brings CDTV development to the same level as any other application, allowing you to deliver the same quality.

SimCD gives you a CDTV within your Amiga, allowing you to iteratively test your application without pressing a lot of CDs. Of course, you should still periodically cut a CD for testing on an actual CDTV. This gives you the final proof and the chance to test your application with a 68000, AmigaDos 1.3, actual memory left over from a boot, and the real CD-ROM drive.

Installation

We recommend using the new methodology of local directories in the standard system directories. In this way you can keep all of the non standard files separate from the standard AmigaDos files. For SimCD, create a "Local" directory in LIBS:, DEVS:, and L:; and add the following lines to your S:User-Startup file:

```
assign  LIBS:  LIBS:Local  add
assign  DEVS:  DEVS:Local  add
assign  L:     L:Local     add
```

The following files represent SimCD. Copy them to the appropriate directories.

<u>File</u>	<u>Destination Directory</u>
Devs/Local/bookmark.device	DEVS:Local
Devs/Local/cdtv.device	DEVS:Local
L/Local/CDFS	L:Local
Libs/Local/debox.library	LIBS:Local
Libs/Local/playerprefs.library	LIBS:Local
SimCD	Where programs normally go (BIN:, etc.)
BookFile	
FakIR	

Support for Simulator

SimCD consists of the following modules: SimCD, FakIR, BookFile, cdtv.device, bookmark.device, CDFS, playerprefs.library, and debox.library. Each can be used separately, though SimCD provides the primary control.

FakIR

FakIR simulates the CDTV Infrared Remote controller with the keyboard. In particular, the PLAY/PAUSE, STOP, FF, REW keys and the joypad are simulated.

FakIR can be run from within SimCD or from the command line. Running FakIR again or "FakIR OFF" will turn off the IR simulation. In addition, FakIR's operation may be suspended and re-enabled by pressing Ctrl, Shift, both Alt keys, and typing the toggle character (default: "i").

When FakIR is active, the arrow keys on the keyboard mimic the IR joypad. The PLAY/PAUSE, STOP, FF, and REW keys are mapped to the following function keys:

<u>IR Key</u>	<u>Function Key</u>
STOP	F1
REW	F2
PLAY/PAUSE	F3
FF	F4

When one of the above function keys is pressed, the rawkey code for the respective IR key is sent. The unusual behavior of the FF and REW keys is also simulated. The simulation aims for a general "feel" so that you can perfect your user interface.

Batch

USAGE: FakIR [options] [ON/OFF]

-t<c> toggle character (default 'i'), example -tc

To toggle the IR simulator on or off while running, hold down Ctrl, left Shift, both Alts, and the toggle character.

To remove the simulator, simply run this program again.

Options

[ON/OFF]

Force FakIR on or off. Useful in batch files and the like.

-t<c>

Change toggle character to <c>, useful in hand dexterity contests.

General Errors

“Cannot open console.device”

“Cannot install input handler”

These errors should never occur.

BookFile

SimCD allows you to load files into bookmark or cardmark memory and will create cardmark memory if not available. BookFile is provided so that you can retrieve information from a running CDTV as well as under the simulation. It provides more capability, allowing you to set up multiple test conditions, etc.

Batch

USAGE: BookFile [options] <file>

-l load memory from file

-s save memory to file

-c use cardmark instead of bookmark

-f create false cardmark memory for simulation
(implies -l (load) and -c (cardmark))

Options

<file>

File to be loaded or saved.

-l

Load file into bookmark or cardmark memory. This will copy the file exactly into the memory and thus must be the same size.

-s

This will save the memory to a file.

-c

Use cardmark memory instead of bookmark. If cardmark does not exist, BookFile will not be able to handle the request.

-f

Use for the initial creation of cardmark memory. The cardmark memory size is determined by the size of the file. -c and -l options are implied by this option. This option should only be used once, otherwise multiple cardmark devices are created.

Examples

Assuming you have created a particular set of bookmarks with your application on a CDTV and would like to use this set in you testing, use BookFile on CDTV as follows:

```
BookFile -s df0:Book.tst
```

Let us say that you a given set of cardmarks, such as special application data, and you wish to always start testing with this same set. Use BookFile the first time as follows:

```
BookFile -f Card.tst
```

And the following times as:

```
BookFile -c -l Card.tst
```

General Errors

“Cannot create StdIO”

BookFile could not create an IO request for bookmark.device or cardmark.device.

“Could not create false cardmark memory”

BookFile was not able to create cardmark memory, probably not enough memory, the file was too large, or the file was not the right file.

“Cannot open device”

BookFile could not open bookmark.device or cardmark.device.

“Not enough memory”

BookFile needs 32K for buffers to handle the bookmark or cardmark devices.

“Cannot open file”

BookFile could not open the bookmark or cardmark files.

“I/O error sizing memory”

Bookmark.device or cardmark.device may be corrupt, they were not reporting sizes correctly.

“File size does not match memory size”

BookFile currently only supports dump files that are exact copies of the bookmark or cardmark memory. If the file is a different size, then they do not match.

“Cannot reformat memory”

“Cannot load memory”

The bookmark or cardmark device could not handle the information. They might be corrupt. Remember that both devices and their information are in the regular memory area and can be affected by any application running. Make sure you use enforcer and mungwall when testing your applications.

“Cannot dump memory to file”

BookFile could not get the memory from the bookmark or cardmark. They might be corrupt, see above.

“Cannot write memory to file”

BookFile could not write to the file.

DEVS:cdtv.device

Cdtv.device is basically the same device driver used on CDTV, except for being AmigaDOS loadable, the ability to use scsi.device and thus a hard drive partition, and seek timing simulation. It is used when SimCD sets up the simulation of CD0: with CDFS, but it can also be used separately for other partitions by creating your own mountlist entry. It is assumed that in doing this you are fully aware of how to set up a mountlist entry. If you are not familiar with this, please get some help - this is the quickest way to grunch your entire system.

In any case, your mountlist must contain CDFS as the file system, since it is needed to interpret the ISO-9660 image that ISOCD will place there. An example of one follows:

```
CDX:
  Device          = cdtv.device
  Filesystem      = L:CDFS
  Unit            = 6
  Flags           = 0
  Surfaces        = 1
  BlocksPerTrack  = 553
  Interleave      = 0
  Reserved        = 2
  LowCyl          = 2
  HighCyl         = 637
  Buffers         = 2
  BufMemType      = 1
  Mount           = 1
  GlobVec         = -1
  StackSize       = 4096
  DosType         = 0x43443031
#
```

Once loaded into memory, cdtv.device cannot be expunged. It must be placed in the DEVS: directory, preferably in the DEVS:Local directory. Refer to the CDTV documentation on cdtv.device for information on dealing with it directly.

Commands

These are the commands documented in the cdtv.device autodocs and notation on what is simulated.

<u>Command</u>	<u>Simulation</u>	<u>Notes</u>
CDTV_ADDCHANGEINT	Not Useful	
CDTV_CHANGENUM	Not Useful	
CDTV_CHANGESTATE	Not Useful	
CDTV_FADE	Simulated	No actual audio
CDTV_FRAMECALL	Simulated	75Hz timing!
CDTV_FRAMECOUNT	Simulated	
CDTV_FRONTANEL	Ignored	
CDTV_GENLOCK	Ignored	
CDTV_INFO	Ignored	
CDTV_ISROM	Simulated	Needs the TOC file
CDTV_MOTOR	Not Useful	
CDTV_MUTE	Simulated	No actual audio
CDTV_PAUSE	Simulated	
CDTV_PLAYLSN	Simulated	No actual audio, needs TOC
CDTV_PLAYMSF	Simulated	No actual audio, needs TOC
CDTV_PLAYSEGSLSN	Simulated	No actual audio, needs TOC
CDTV_PLAYSEGMSF	Simulated	No actual audio, needs TOC
CDTV_PLAYTRACK	Simulated	No actual audio, needs TOC
CDTV_POKEPLAYLSN	Simulated	No actual audio, needs TOC
CDTV_POKEPLAYMSF	Simulated	No actual audio, needs TOC
CDTV_POKESEGLSN	Simulated	No actual audio, needs TOC
CDTV_POKESEGMSF	Simulated	No actual audio, needs TOC
CDTV_QUICKSTATUS	Simulated	
CDTV_READ	Simulated	
CDTV_READXL	Simulated	
CDTV_REMCHANGEINT	Not Useful	
CDTV_RESET	Ignored	
CDTV_SEEK	Simulated	
CDTV_STOPPLAY	Simulated	
CDTV_SUBQLSN	Not Useful	
CDTV_SUBQMSF	Not Useful	
CDTV_TOCLSN	Simulated	No actual audio, needs TOC
CDTV_TOCMSF	Simulated	No actual audio, needs TOC
CDTV_WRITE	Fully Simulated	Not!

DEVS:bookmark.device

Bookmark.device controls bookmark and cardmark memory. Make sure that it is in the DEVS: directory, preferably in the DEVS:Local directory. Like cdtv.device, it is not expungable, since the memory it deals with is supposed to hang around. Refer to BookFile above and the CDTV documentation on direct programming of the devices.

L:CDFS

CDFS is the file system that allows you to use an ISO-9660 image as a normal AmigaDos volume. It is set up automatically for you through SimCD to simulate CD0:. It must be in the L: directory, like the others, preferably in the L:Local directory. When you use it in your own mountlists, you can use cdtv.device or scsi.device as the device for the partition. If you were to use scsi.device, CDFS would use the partition at full speed, without seek timing simulation or any cdtv.device code. This is useful when you wish to use an ISO-9660 partition for other than simulation. Since this format is read only, there is no need for any of the dynamic file handling. Files are always in one piece, no fragmentation or scattered layout of the files.

To use independent of SimCD, create a mountlist entry for the partition that will hold the image or use HDToolBox to set up a partition. If you use HDToolBox, follow these steps:

- Make sure that SimCD is installed on the system, since HDToolBox will look for CDFS.
- Start up HDToolBox.
- Select the drive where you want the ISO-9660 partition.
- Click on "Partition Drive", this will take you to the Partitioning control panel.
- Select or create the partition to contain the image. Be sure to make it large enough, at least as large as all of your files and try to have room for the final application size.

You can use an entire drive for the simulation or just a part of it, sharing the drive with the source material, for example.

Do *not* chose "CD0:" as the name, SimCD uses that as the simulated drive name.

- Click on "Advanced Options" to gain access to the file systems settings.
- Click on "Add/Update File Systems", this will take you to the File System Maintenance control panel.

- Click on "Add New File System".
 - In the requester that appears, enter the following:
 - In the Filename, enter "L:CDFS."
 - In the DosType, enter "0x43443031."
 - In the Version, enter "25."
 - Click on "OK."
- The custom file system should appear in the list, click on "OK."
- Click on "Change File System for Partition," this will take you to the File System Characteristics panel.
 - Click on "Custom File System," this will also enable the "Identifier" string gadget.
 - Change the text in the "Identifier" string gadget to "0x43443031."
 - Click on "OK."
 - Click on "OK" again to return to the main control panel.
- Click on "Save Changes to Drive." This will write the new RigidDiskBlock out to the drive with the CDFS file system on it. Be sure to do this before rebooting!
- Click on "Exit," HDToolBox should ask you to reboot, do so now.

The CDFS partition should now be available to AmigaDos. Now it is ready for ISOCD to write the ISO-9660 image to it. If a new version of CDFS is ever released, you will need to use HDToolBox again to update the custom file system.

Until ISOCD places a valid ISO-9660 image on the partition, you will notice long pauses at boot time. This is because CDFS is trying to find a valid PVD, or Primary Volume Descriptor, on the image. It will read 256 sectors in its attempt to find a valid PVD. This will cause a long pause, as if the drive was being validated, but you do not need to worry. It will never be seen again, once you have an ISO-9660 image on the partition.

Note: This version of CDFS retains the date error present in CDTV V1.00 that does not report ISO-9660 dates properly. It will show a date one day later than is correct in most cases.

LIBS:playerprefs.library and LIBS:debox.library

Playerprefs.library and debox.library are provided and should be placed in LIBS:, preferably the LIBS:Local directory. Refer to the CDTV documentation on the usage of these two libraries. Bookit is usable to set the preferences usage, refer to the bookit documentation. Keep in mind that playerprefs.library is 170K and if you invoke the bouncing logo screen blanker, even more memory is needed.

Memory/Resource Usage

SimCD itself uses little resources, but does provide a guess as to the probable memory usage of the options chosen to be simulated. Also, SimCD requires AmigaDos 2.0 or higher.

Once Simulator is Running

Once the simulator is running and CD0: is available, you can still run SimCD to change some of the options. CDFS and the three device drivers, cdtv, bookmark, and cardmark, are not removable once you have started them. If CDFS/cdrv.device is chosen, Sim Device and TOC files cannot be changed. If bookmark or cardmark is chosen, the corresponding files cannot be changed.

Files

Settings

If no settings file is chosen, SimCD will use S:SimCD.Default. Load and save options are available from the menus. Use multiple files for separate applications or test configurations.

Sim Device

The Sim Device specifies the hard disk partition that is to be used for CD0:. It must have been created with either a mountlist entry or through HDToolBox, see CDFS under Settings for information on creating a partition. The partition can be one that is already controlled by CDFS without cdrv.device. Remember to not enter CD0:, since CD0: will be the drive that results from the simulation. The Sim Device is the same one that ISOCD uses to write to as the Image drive. This option is unchangeable once used.

Book Mark

Book Mark refers to the file loaded into the bookmark memory. You can use the BookFile program to capture bookmark memory from CDTV, refer to the BookFile section under Support for Simulator. Like Sim Device, this option is unchangeable once used.

Card Mark

Just like Book Mark, this specifies the file for cardmark memory. If there is no cardmark memory, it will be created. The BookFile program also handles cardmark memory. Card Mark is also unchangeable once used.

TOC

The TOC file specifies the CD Table of Contents for cdtv.device to simulate. It specifies the types and sizes of the various tracks to be simulated. The sizes are in frames (sectors). Specify Data or Audio for the type, a tab or spaces, then the size.

ISOCDB will provide you with the size of the data track or ISO-9660 image.

For example, if the ISO-9660 image used 64 megabytes and was followed by three audio tracks, the TOC file would contain:

```
Data  32768
Audio 15750
Audio 45322
Audio 4500
```

Cdtv.device uses this information for replying to the TOC commands. This file is also unchangeable once CD0: is simulated.

Settings

These are the settings for the simulation. CDFS and cdtv.device are necessary for simulation of CD0:. A lot of these options will remain on after simulation until a reboot. The two libraries can be purged as long as they are not currently opened; IR and No Fast Memory are always available to be turned on or off.

CDFS

CDFS is necessary for simulation of CD0:. You need to set the Sim Device to the partition that you have set aside as the ISO-9660 volume. You also need to create this partition first with ISOCDB before simulating it. Cdtv.device is necessary if this option is used. After simulation you can find the file MountCDTV.tmp in the T: directory. It is the mountlist entry that mounted CD0: if you wish to see how it was mounted.

cdtv.device

Cdtv.device will use the TOC file if you specify it, but it is not necessary. The TOC file would be needed if you have any audio tracks used in your application. Refer to TOC above.

bookmark.device

The bookmark.device will load the Book Mark file if specified, otherwise it will setup the bookmark memory area blank. This device is needed if you are going to use cardmark.device, since bookmark.device is used to create it. BookFile is available for further management of bookmark.device.

cardmark.device

The cardmark.device requires the Card Mark file and bookmark.device set. The Card Mark file will be loaded into card mark memory. This memory is not recoverable.

playerprefs.library

The playerprefs.library handles CDTV preferences and audio control. Of course, the audio panel is not usable, but you should use the preferences.

debox.library

Debox.library is available for the decompression routines. Both playerprefs and debox are documented in the CDTV documentation.

IR

IR turns the Infared Remote simulation on or off. Refer to FakIR under Support for Simulator.

No Fast Memory

No Fast Memory just calls NoFastMem. Since CDTV has no fast memory, this option is placed here for convenience.

Speed

Speed is available for future use.

Avail

Avail reports an approximation of available memory after choosing Simulate.

Simulate

Simulate starts the requested devices, loads the appropriate memories, etc. After everything is started, SimCD will save the current settings to the Settings file and exit to the shell. If you do not want to simulate, simply close the window, chose Quit from the menu, or hit Amiga-Q.

Load/Save Settings

Both load and save are available from the menus for the settings file. Use these for creating an alternate settings file for a specific application. Then use SimCD for future simulations as:

SimCD -b <file>

Batch

USAGE: SimCD [options] [<Configuration file>]

-b batch execute (do not use interface) (requires Configuration file)

Options

<Configuration file>

This is the file that contains the SimCD settings. It is normally S:SimCD.Default if not specified on the command line.

-b

This will configure the Simulator based on the configuration file. If one is not specified, it will use S:SimCD.Default. This the common usage of SimCD once the settings are chosen for normal usage.

General Errors

“Cannot open settings file”

Configuration file is not available or is corrupted. Remove S:SimCD.Default, run SimCD fresh, and reset all of your settings.

“Intuition (V2.0) not available”

SimCD requires v2.0 of intuition or greater. If you specify -b, SimCD will not use the interface and will not require v2.0.

“Interface not available”

Probably not enough memory for window.

“Device <dev:> not available”

Dev: specified in Sim Device was not found. Check that it is properly set up with HdToolBox or that your mountlist entry is correct. See CDFS under Settings.

“TOC file <file> not available”

SimCD could not open the TOC file.

“More than 99 entries in TOC file”

Just like a CD, SimCD can only handle up to 99 tracks, data or audio. This error message may also occur if you attempt to load an invalid TOC file, the TOC file interpreter is fairly simple.

“TOC file error, line #n”

Text in TOC file was not of the form “Data 32768” or “Audio 45322” in line n. Make sure that you are preparing the TOC file with a text or word processor that saves out text files. Do not use just any file.

“Cannot create StdIO”

This error should not occur.

“Cannot open device”

SimCD could not open bookmark.device or cardmark.device. Check that SimCD has been properly installed and these devices are in DEVS: or DEVS:Local. Check Installation.

“Not enough memory”

SimCD needs 32K for buffers to handle the bookmark or cardmark devices.

“Cannot open file”

SimCD could not open the bookmark or cardmark files.

“I/O error sizing memory”

Bookmark.device or cardmark.device may be corrupt, they were not reporting their sizes correctly.

“File size does not match memory size”

SimCD currently supports only dump files that are exact copies of the bookmark or cardmark memory. If the file is a different size, then they do not match.

“Cannot reformat memory”

“Cannot load memory”

The bookmark or cardmark device could not handle the information. They might be corrupt. Remember that both devices and their information are in the regular memory area and can be affected by any application running.

Glossary

CD0:

The CD-ROM drive on a CDTV or the partition simulated as this drive with SimCD.

CDFS

The CDTV CD File System. This allows you to treat CD0: as just another Amiga drive, even though it is in ISO-9660 format.

cdtv.device

The device that controls the CDTV hardware or the simulation of it with SimCD.

CDXL

The fastest way to transfer data from the CD-ROM drive into the CDTV. CDXL basically starts a read and begins transferring the incoming data into places in memory that you specify. CDXL nodes control the shotgun effect of the data being read in, since the read does not pause. CDXL removes the liability of the CD-ROM seek times, except for the initial seek.

ISO-9660

Refers to the 1988 ISO-9660 standard, specifically ISO 9660: 1988 (E), corrected 1988-09-01. This is the standard produced by ISOCD.

Credits

Documentation by:

Kenneth Yeast
Carl Sassenrath [Previous documentation]

Software by:

Kenneth Yeast
Carl Sassenrath [Author of ROM software, Producer, and "Father" of CDTV]

Also a special thanks to Leo L. Schwab, and everyone at Silent Software for their work on CDTV.

